

Generation Interconnection Request System Impact Study Report

For Queue Project Q14-0303

Mira Loma 230 kV
June 15, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0303. The Interconnection Customer proposes a 279.8 MW wind facility interconnecting at the Mira Loma 230 kV substation in the Illinois area. The request was received on January 17, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 189 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,022,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of EASY ROAD-MIRA LOMA 500.0 kV, the YKK USA CHESTNEY-MIRA LOMA 230.0 kV loads to 113.5% of its 1,044 MVA rating. Project DFAX: 89.7%.

The FISHKILL PLAINS-DILLARD-ORANGE 345.0 kV is overloaded to 101.9% of its 1,913 MVA emergency rating for the loss of the MUSKINGUM RIVER-TIDD 500.0 kV. The project contributes 5.8% DFAX (16.3 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
YKK USA CHESTNEY-MIRA LOMA 230.0 kV	Rebuild line with higher-capacity conductor	\$38,150,000	21% (\$8,159,000)
FISHKILL PLAINS-DILLARD-ORANGE 345.0 kV	Reconductor line	\$49,020,000	4% (\$1,863,000)
	Total allocated to project		\$10,022,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.